

Prospective Germline Exome and Machine Learning-Based Risk Score Identify Predictive and Prognostic Biomarkers of Immunotherapy Outcomes in Advanced Non-Small Cell Lung Cancer

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Abstract

Background: Immune checkpoint inhibitors (ICIs) have transformed the treatment landscape of advanced non-small cell lung cancer (NSCLC). However, a substantial proportion of patients do not experience durable clinical benefits, and established tumour biomarkers, such as PD-L1 expression and tumour mutational burden, often show limited predictive value. The potential of inherited germline variants to predict immunotherapy outcomes in NSCLC remains a critical and underexplored area.

Methods: We prospectively enrolled 117 patients with advanced NSCLC treated with ICI-based regimens at two centres in Spain. Germline whole-exome sequencing (WES) was performed on pretreatment blood samples. Exonic and intronic variants were annotated and integrated with comprehensive clinical data. We applied XGBoost and LASSO machine learning models to identify predictive germline variants and clinical features, and subsequently trained them to predict treatment response and progression-free survival (PFS). This approach produced a novel clinical–germline risk score, generating both a global model and a specific model for the lung adenocarcinoma (LUAD) histological subtype.

Results: XGBoost significantly outperformed penalised regression (LASSO), achieving a robust cross-validated area under the curve (AUC) of 0.845 for predicting treatment response in the validation cohort. Our models identified several novel germline loci that were significantly associated with immunotherapy outcomes. Variants in *SLC6A16*, *SIGLEC11*, *PDE4E*, and *OR10H5* were associated with reduced PFS, whereas variants in *CCZ1B* and *PHLDB1* were associated with extended PFS. Lymph node metastasis was confirmed as the sole independent clinical predictor of poor response (OR 2.07, $P=0.008$). A predictive algorithm that included these individual variables generated a clinical–germline risk score that successfully stratified patients into high- and low-risk groups with markedly different median PFS (low-risk: 18 months vs. high-risk: 7 months, log-rank $P < 0.001$), retaining discriminatory power across histological subgroups.

Conclusions: The integration of specific germline variants as the output of advanced machine learning analysis of the exome with key clinical features provides accurate and novel predictive information for immunotherapy in NSCLC. This approach not only uncovers new genetic biomarkers but also supports the clinical adoption of composite risk scores for personalised precision immunotherapy, paving the way for improved patient selection and stratification.

Background

Non-small cell lung cancer (NSCLC) represents approximately 85% of all lung cancer cases and continues to be the leading cause of cancer-related mortality worldwide [1]. Immune checkpoint inhibitors (ICIs), notably those targeting the PD-1 and PD-L1 pathways, have transformed the treatment paradigms for advanced NSCLC and significantly extended survival in select patients [2, 3]. However, only a subset of individuals achieves durable clinical responses, emphasising the need for robust biomarkers to guide optimal patient selection.

Currently used tumour-based predictors, such as PD-L1 expression, tumour mutational burden, and microsatellite instability, show limited and inconsistent values across real-world populations [4, 5]. Consequently, there is an urgent need to expand the biomarker repertoire beyond the tumour

microenvironment. Recently, host-related factors, specifically germline genetic variations, have gained recognition as critical modulators of anti-tumour immune responses, influencing both treatment efficacy and immune-related toxicities [6, 7]. Several retrospective and multi-omics studies have linked germline variants to immune phenotypes and clinical endpoints in ICI-treated NSCLC cohorts [8–10]. However, germline-informed predictive models have seldom been prospectively validated or integrated with clinical and tumour-derived data within real-world settings, limiting their clinical translation.

Concurrently, advances in machine learning (ML) and artificial intelligence (AI) have enabled the analysis of complex multidimensional clinicogenomic datasets, uncovering subtle and non-linear predictors of therapeutic efficacy that escape traditional statistical methods[11–14]. Despite this technological progress, the translation of germline genomics into clinically actionable decision-making tools remains unfulfilled.

Building on current evidence, we hypothesised that specific inherited variants are significantly associated with differential responses to ICIs in patients with advanced NSCLC and that machine learning-based integration of these signals facilitates robust risk stratification.

Here, we present a prospective, multicentre study that systematically integrates whole-exome germline sequencing, comprehensive clinical data, and state-of-the-art ML algorithms to develop and validate a composite risk score for immunotherapy benefit in NSCLC, aiming to advance biomarker-driven precision medicine.

Methods

Study Design and Patient Cohort

This prospective, multicentre study enrolled 117 adult patients (≥ 18 years) with histologically confirmed stage IIIB/IV NSCLC treated between January 2018 and June 2024 at two hospitals in Málaga (Hospital Universitario Virgen de la Victoria and Hospital Regional Universitario de Málaga), Spain. Eligible patients were scheduled to receive immune checkpoint inhibitor (ICI)-based therapy, either as monotherapy or in combination with chemotherapy. Key exclusion criteria included prior ICI exposure and a history of other malignancies (within five years, except for non-melanoma skin cancer). All patients provided written informed consent. The study was approved by the regional ethics committee and conducted in accordance with the Declaration of Helsinki. To ensure patient privacy during machine learning analysis, all clinical and genomic data were pseudonymized and stored on secure servers, in strict compliance with the General Data Protection Regulation (GDPR) (EU) 2016/679

Sample Collection and Clinical Data

Peripheral blood samples were collected in EDTA tubes before the first ICI administration. Germline DNA was extracted using the QIAamp DNA Mini Kit (Qiagen) and quantified with Qubit dsDNA BR Assay Kit (Thermo Fisher Scientific). Comprehensive clinical data were collected from electronic medical records, including age, sex, histological subtype (adenocarcinoma [LUAD], squamous cell carcinoma [LUSC], or others), TNM stage (IIIB vs. IV), treatment regimen, PD-L1 expression status, metastatic sites, and ECOG performance status. Treatment efficacy was evaluated according to the RECIST v1.1 criteria [15], with tumour assessment

performed every 8–12 weeks according to clinical practice. Clinical outcomes were categorised as Complete Response (CR), Partial Response (PR), Stable Disease (SD), or Progressive Disease (PD). [15]

Whole-Exome Library Preparation and Sequencing

Libraries were prepared from 50 ng of germline DNA using the Twist Human Core Exome 2.0 Kit (Twist Bioscience), which targets a total of 33 Mb of coding regions. The quality of the libraries was assessed using an Agilent 2100 Bioanalyzer. Sequencing was performed on an Illumina NovaSeq 6000 platform (2 × 150 bp), achieving a mean coverage of > 30x. FastQC v0.11.9 [16] and Fastp v0.39. [17] were used for raw data quality control and adapter trimming, respectively.

Bioinformatics Processing and Variant Annotation

Sequencing reads were aligned to the GRCh38 human reference genome using the BWA-MEM. Duplicates were marked, and base quality scores were recalibrated using Picard and GATK. Germline variants were called with GATK HaplotypeCaller, filtered per GATK best practices [18] (minimum depth > 30x, QD > 2.0, FS < 60.0, MQ > 40.0), and restricted to the 40–60% allelic fraction to minimise clonal haematopoietic artefacts. Functional annotation was performed using CRAVAT [19] and gnomAD v3.1.2 [20] for allele frequency analysis.

Variants were classified according to pathogenicity, following The American College of Medical Genetics and Genomics (ACMG) recommendations [21]. Only rare (MAF < 1%), exonic, splice-site, and high-impact intronic variants were selected. To reduce data redundancy and minimise multicollinearity, we applied a pairwise correlation filter; for variant pairs with high correlation (Pearson's $r \geq 0.70$), only one representative variant was retained for downstream analysis. Variant annotation was blinded to the clinical outcomes.

Statistical Analysis

Associations between genetic variants and clinical benefits were assessed using Fisher's exact test and logistic regression with FDR correction. To align with the concept of disease control, the outcome variable was dichotomised into responders (patients achieving CR, PR, or SD) and non-responders (patients with PD). Progression-free survival (PFS) and overall survival (OS) were compared using Kaplan–Meier curves and log-rank tests. Analyses were run in R v4.2.0 (packages: tidyverse, survival, xgboost). Statistical significance was set at $P < 0.05$.

Machine Learning and Predictive Modelling

To ensure robust phenotypic separation, we first defined a cohort of 'extreme responders' (non-responders < 3 months vs. responders > 12 months). Using this cohort, we considered four modelling scenarios: (1) all selected germline variants, (2) exonic-only variants, (3) all variants per LUAD subgroup, and (4) exonic variants of LUAD only.

For model construction, the dataset corresponding to these extreme phenotypes was split into training (70%) and test (30%) sets, and 1,000 iterations were performed to evaluate model stability and feature importance. Within each training set, hyperparameter tuning used stratified 5-fold cross-validation to minimise data leakage.

To minimise overfitting and ensure rigorous evaluation, model construction was performed on a training set comprising 70% of the extreme responder population. For internal validation, we employed a temporal hold-out strategy, reserving the subset of extreme responder patients recruited during the final 12 months of the study period ($n = 37$). This temporal split mimics a prospective clinical scenario in which the model is tested on future patients, providing a more robust estimate of predictive performance than random splitting.

The optimised model from each iteration was evaluated on the test set to assess predictive performance (AUC). The AUC values reflect the predictive accuracy of the immunotherapy response within the held-out test sets to ensure generalisability. Feature selection, model stability, and variable importance metrics were collated across the iterations. Subsequent survival modelling grouped patients according to genotype status (variant carriers vs. non-carriers) and compared outcomes using the log-rank test. To define the optimal thresholds, the `surv_cutpoint` function from the *Survminer* R package was used to identify the cutoff value that maximised the log-rank statistic. Based on the biological premise that mechanisms facilitating initial immune recognition (response) are essential for durable disease control, we hypothesised that robust response predictors would also stratify survival outcomes. Consequently, two composite risk scores were generated by integrating variables that demonstrated consistency across both the ML response model and univariate PFS analyses: (i) The Global Clinical-Germline Score (all patients), comprising six adverse predictors (five molecular, one clinical) and three favourable predictors (two molecular, one clinical), and (ii) The LUAD-Specific Score (adenocarcinoma only), comprising three adverse predictors and three favourable predictors. Adverse predictors ($OR > 1$) were incorporated as penalties by adding their $\log(OR)$ values, whereas favourable predictors ($OR < 1$) were treated as protective factors by subtracting $\log(OR)$ values. This weighting strategy reflects the magnitude of the association between each variable and the clinical outcome. The thresholds for the high- and low-risk groups were determined using an optimal cutoff based on (a) a minimum subgroup size of 10% and (b) maximisation of survival curve separation. To assess threshold robustness, a nonparametric bootstrap procedure (1,000 resamples) was used to estimate the 95% confidence interval of the optimal cutoff distribution.

External Validation Cohorts

To evaluate the robustness and generalisability of the derived clinical-germline risk score, we analysed two independent validation cohorts. First, to assess the performance of the gene signature in an external NSCLC population and given the scarcity of public datasets with matched germline-clinical immunotherapy data, we queried cBioPortal for Cancer Genomics. We selected the dataset from Rizvi et al. 2015/Hellmann et al. 2018, comprising 75 patients treated with immune checkpoint inhibitors [22]. As this dataset primarily provides somatic sequencing data, validation was restricted to assessing the prognostic value of the identified gene signature at the gene level. Second, to explore the utility of the score across different tumour types (cross-tumour validation), we included an independent in-house cohort of 19 patients with metastatic melanoma. These patients were treated with anti-PD-1/PD-L1 monotherapy at Hospital Regional de Malaga, following the

same inclusion criteria and ethical standards as those of the main cohort. Germline sequencing and clinical data collection for this subset followed the pipeline described above.

Data and Code Availability

Raw sequencing data, genotype matrices, and scripts can be obtained from the corresponding authors upon reasonable request and institutional approval.

Results

Patients' characteristics

A total of 117 patients with advanced NSCLC were enrolled. The median age was 67.5 years (range: 46–85 years), and 74.4% were male. Histologically, lung adenocarcinoma (LUAD) accounted for 59.8% of the cases, while 30.8% were squamous cell carcinomas (LUSC). Although 19.1% of patients were initially diagnosed with locally advanced disease (Stage IIIB), all presented with metastatic progression or Stage IV disease at the time of ICI initiation. Among those with metastatic spread, the most frequent sites were the lungs (51.3%) and the lymph nodes (47.9%). ICI monotherapy was administered in 25.6% of cases, with combination chemo-ICI or other regimens administered in 74.4% of cases. At the first evaluation per RECIST v1.1, the Objective Response Rate (CR or PR) was 17.1% in evaluable patients, whereas 65% achieved Disease Control (CR, PR, or SD) by 3 months. The detailed cohort characteristics are presented in Table 1.

Table 1

Demographic and Clinical Features of the Study Population Summary of the baseline characteristics of the 117 patients with advanced NSCLC included in the study.

Variable	Category	n (117)	%
Sex	Female (F)	30	25.6
	Male (M)	87	74.4
Histology	LUAD	70	59.8
	LUSC	36	30.8
	NSCLC NS	7	6.0
	Other	4	3.4
Stage at diagnosis	IIIB (locally advanced)	22	19.13
	IV (Metastatic)	83	72.17
PD-L1 expression	< 1%	63	53.8
	1–49%	35	29.9
	> 50%	10	8.5
	Not evaluated	9	7.7
Therapy type	ICI monotherapy	30	25.6
	CT + ICI	43	36.8
	CT + ICI + Others	39	33.3
	Immunotherapy combo	1	0.9
RECIST 1st evaluation (8–12 weeks)	Complete Response (CR)	5	4.3
	Partial Response (PR)	15	12.8
	Stable Disease (SD)	3	2.6
	Progressive Disease (PD)	11	9.4
	Not evaluable	83	70.9
Clinical Benefit At 3 months (Responder vs Non-Responders)	Responders	76	65.0
	Non-responders	41	35.0
Clinical Benefit At 12 months (Responder vs Non-Responders)	Responders	39	33.3

* Abbreviations: CT, chemotherapy; ICI, immune checkpoint inhibitor; IT, immunotherapy; LUAD, lung adenocarcinoma; LUSC, lung squamous cell carcinoma; NSCLC NOS, non-small cell lung cancer not otherwise specified; NE, not evaluable; M1, distant metastasis. "Other" histology includes large cell carcinoma and adenosquamous carcinoma. M1 sites refer to distant metastases confirmed at the baseline.

Variable	Category	n (117)	%
	Non-responders	69	59.0
M1 Metastasis			
	Brain	14	12.0
	Liver	12	10,25
	Bone	40	34,18
	Lung	60	51,28
	Skin	30	25,64
	Lymph node	56	47,86
* Abbreviations: CT, chemotherapy; ICI, immune checkpoint inhibitor; IT, immunotherapy; LUAD, lung adenocarcinoma; LUSC, lung squamous cell carcinoma; NSCLC NOS, non-small cell lung cancer not otherwise specified; NE, not evaluable; M1, distant metastasis. "Other" histology includes large cell carcinoma and adenosquamous carcinoma. M1 sites refer to distant metastases confirmed at the baseline.			

Clinical Variables associate with Immunotherapy Response

Multivariable logistic regression confirmed that lymph node metastasis was the sole independent clinical predictor of poor response to ICIs (OR 7.9, 95% CI 1.8–43.3, $P = 0.008$). Given that all patients were treated in the advanced/metastatic setting, the tumour stage was not included as a covariate. Regarding the metastatic burden, no specific organ involvement (brain, liver, bone, lung, or skin) other than lymph nodes was significantly associated with the outcome, as shown in Fig. 1. Neither treatment modality (ICI monotherapy vs. combination), sex, PD-L1 expression strata, nor ECOG performance status were significantly associated with clinical benefits.

Response Germline Variant Discovery and Model Performance

To identify germline predictors of immunotherapy response, we implemented a comprehensive variant filtering and prioritisation pipeline (Fig. 2a). From an initial pool of 108,944 called variants, strict quality control and linkage disequilibrium pruning (Pearson's $r < 0.70$) yielded a refined set of candidates. Univariate statistical filtering further narrowed this pool to those showing the strongest association with clinical benefit.

The predictive performance of the machine learning models was evaluated across four analytical scenarios. The XGBoost model utilising all exonic variants achieved the highest accuracy, with a cross-validated median area under the ROC curve (AUC) of 0.845 (IQR 0.742–0.879) in the held-out test set (Fig. 2b). Comparative modelling consistently demonstrated the superiority of XGBoost over LASSO regression across all datasets. For the LUAD-specific cohort, XGBoost achieved an AUC of 0.82, compared to 0.56 for LASSO. Similarly, in models restricted to exonic variants within LUAD, XGBoost maintained a robust performance (AUC 0.77) compared to LASSO (AUC 0.72).

Feature importance analysis revealed distinct genetic signatures depending on the histological context. In the global analysis (all patients), the most recurrently selected loci (> 70% of iterations) included variants in *OR51M1*, *C22orf34*, *DENND2B*, *SLC6A16*, *ZNF148*, *SIGLEC11*, *ZNF454*, and *PDE4E* genes. Conversely, the analysis focused specifically on the LUAD subgroup highlighted distinct drivers, identifying *ARAP3*, *CCZ1B*, *ARHGAP22*, and *OR10H5* as significant contributors. The frequent selection of these variants suggests their potential role in germline-mediated modulation of the immune response.

Statistical Validation and Prognostic Characterization of Prioritised Variants

To individually validate the germline features prioritised by the best-performing XGBoost model (exonic variants) and quantify their effect sizes, we performed univariate logistic regression analysis. This post-hoc analysis confirmed that several germline variants were significantly associated ($P < 0.05$) with clinical benefits (responders vs. non-responders). Specifically, variants chr19:49309146 G > T (*SLC6A16*), chr19:49960436_>T (*SIGLEC11*), chr5:178964901 A > C (*ZNF454*), chr5:60460204 A > G (*PDE4E*), chr10:48451594 T > C (*ARHGAP22*), and chr19:15794719 C > A (*OR10H5*) were associated with a higher likelihood of non-response (resistance). In contrast, chr11:8766678 A > C (*DENND2B*), chr11:118643907 C > T (*PHLDB1*), chr5:141671669 C > T (*ARAP3*), and chr7:6822249 G > C (*CCZ1B*) were associated with a positive treatment outcome (Table 2).

Table 2

Prioritised Germline Variants and Statistical Associations with Immunotherapy Outcomes List of the top germline variants selected by machine learning models and univariate logistic regression, showing odds ratios, confidence intervals, and p-values for immunotherapy response, progression-free survival (PFS), and overall survival (OS). Genes, variant positions, variant types (synonymous, non-synonymous, and intronic), and relevant molecular functions were annotated for each locus

Gene	Variant (chr:position/ref/alt)	Variant Type	Univariate			PFS	OS
			OR	95% CI	p-val	Log-Rank P-val	Log-Rank P-val
OR51M1	chr11:5390165:C > A	Non-synonymous	2.94	(0.7–14.5)	0.1	0.89	0.96
DENND2B	chr11:8766678:A > C	Intronic	0.28	(0.08–0.86)	0.03*	0.067	0.22
SLC6A16	chr19:49309146:G > T	Intronic	4.90	(1.73–15.8)	0.004**	0.0048**	0.0057**
SIGLEC11	chr19:49960436:_.T	Intronic	8.76	(2.5–41)	0.002**	0.03*	0.00021**
C22orf34	chr22:49624397:G > A	Intronic	3.74	(0.79–26.)	0.1	0.15	0.043*
ZNF148	chr3:125232977:C > T	Non-synonymous	2.14	(0.58–8.9)	0.3	0.96	0.84
ZNF454	chr5:178964901:A > C	Non-synonymous	4.71	(1.28–22.6)	0.03*	0.34	0.17
PDE4E	chr5:60460204:A > G	Intronic	6.13	(2.5–21.1)	0.002**	0.0051**	0.0001**
PHLDB1	chr11:118643907:C > T	Non-synonymous	0.15	(0.04–0.44)	0.001**	0.036*	0.011*
ARAP3	chr5:141671669:C > T	Non-synonymous	0.22	(0.06–0.63)	0.007**	0.085	0.0001**
ARHGAP22	chr10:48451594:T > C	Intronic	3.64	(1.32–10.5)	0.01**	0.049*	0.05*
CCZ1B	chr7:6822249:G > C	Intronic	0.14	(0.04–0.42)	0.001**	0.0016**	0.00031**
OR10H5	chr19:15794719:C > A	Non-synonymous	7.03	(2.03–33.1)	0.005**	0.0075**	0.028*

Furthermore, we assessed progression-free survival (PFS) to corroborate these findings. The median PFS was compared between patients harbouring the variant (variant-positive) and those with the reference genotype (wild type). Consistent with the logistic regression results, the *CCZ1B* variant (chr7:6822249 G > C) was associated with significantly extended PFS (21 vs. 9 months; $P = 0.0016$), aligning with its protective odds ratio OR (0.14). A similar survival benefit was observed for *PHLDB1* (chr11:118643907 C > T; 12 vs. 8 months; $P = 0.036$).

Conversely, the presence of variants in *SLC6A16* (chr19:49309146 G > T) ($P = 0.0048$), *OR10H5* (chr19:15794719 C > A) ($P = 0.0075$), *PDE4E* (chr5:60460204 A > G) ($P = 0.0051$), *ARHGAP22* (chr10:48451594 T > C) ($P = 0.049$), and *SIGLEC11* (chr19:49960436_>T) ($P = 0.03$) was associated with significantly reduced PFS compared to non-carriers. Variants in *DENND2B* (chr11:8766678 A > C) and *ARAP3* (chr5:141671669 C > T) showed a trend toward PFS benefits ($P = 0.067$ and $P = 0.085$, respectively), although the differences were not significant.

Finally, we evaluated the Overall Survival (OS). Most notably, the *CCZ1B* (chr7:6822249 G > C) variant conferred a substantial survival advantage (median OS, 22 vs. 7 months; $P < 0.01$). While *PHLDB1* (chr11:118643907 C > T) and *ARAP3* (chr5:141671669 C > T) variants achieved significance for OS (despite missing the PFS threshold), variants in *PDE4E* (chr5:60460204 A > G), *SLC6A16* (chr19:49309146 G > T), *SIGLEC11* (chr19:49960436_>T), *ARHGAP22* (chr10:48451594 T > C), and *OR10H5* (chr19:15794719 C > A) were associated with a marked decline in OS. Interestingly, the *C22orf34* variant (chr22:49624397 G > A), which was not significant in the response or PFS analyses, was identified as a novel predictor of poor OS ($P = 0.043$). The representative Kaplan–Meier curves are shown in Fig. 4.

Multivariable Analysis of Prioritized Germline and Clinical Data

To determine the independent predictive value of the identified biomarkers, we conducted a comprehensive multivariable logistic regression analysis. This model integrated the germline variants selected by the XGBoost algorithm, specifically those showing consistency in the univariate response and progression-free survival (PFS) assessments, along with key clinical covariates (sex, histology, PD-L1 expression, treatment type, and metastatic sites). In all model iterations, lymph node metastasis consistently emerged as a robust independent predictor of resistance, yielding Odds Ratios (OR) for non-response ranging from approximately 5.9 to 21.0 ($P < 0.05$) depending on the variant combination (Fig. 5). Regarding germline features, six variants maintained a significant correlation with treatment outcomes after adjusting for clinical factors. Variants in *SLC6A16* (chr19:49309146 G > T), *PDE4E* (chr5:60460204 A > G), and *ARHGAP22* (chr10:48451594 T > C) were confirmed as independent predictors of poor response. Patients carrying these variants experienced a significantly higher likelihood of resistance and shorter median PFS.

Conversely, the *PHLDB1* (chr11:118643907 C > T), *CCZ1B* (chr7:6822249 G > C), and *OR10H5* (chr19:15794719 C > A) variants were independently associated with favourable outcomes, exhibiting protective odds ratios in the regression model, and correlated with extended median PFS in carriers (Fig. 5).

No consistent independent associations were found for other clinical factors, including distant metastases, immunotherapy regimen, sex, PD-L1 expression, or ECOG status. Consequently, lymph node involvement and the identified germline profile were the most reliable independent indicators of the clinical benefit. Notably, these predictive effects were observed for both non-synonymous and intronic variants, suggesting diverse functional mechanisms of immune modulation.

Development and Validation of the Clinical-Germline Prognostic Score

Finally, to translate these findings into a clinical tool, we constructed a composite risk score by integrating the prioritised germline variants with lymph node status, weighting each variable by its corresponding regression

coefficient (log-odds ratio). Patients were stratified into high- and low-risk groups based on the optimal cutoff determined by the `surv_cutpoint` algorithm.

Kaplan–Meier analysis demonstrated a robust distinction in clinical outcomes between the strata. In the overall cohort, individuals in the low-risk category experienced significantly longer progression-free survival (PFS) than those in the high-risk group (median 18.0 vs. 7.0 months; log-rank $P < 0.001$) (Fig. 6a). This stratification power was maintained in the LUAD subgroup (median PFS 19.0 vs. 5.0 months; $P = 0.012$) (Fig. 6b).

Validation in Independent Cohorts

To assess the robustness of the score, we first applied a temporal hold-out strategy and tested the model on a subset of patients recruited during the final study period ($n = 22$). Using the predefined cutoff, the score successfully stratified PFS ($P = 0.002$; median 18 vs. 4 months), confirming internal stability (Fig. 7a-b).

Subsequently, we explored the score's utility in external settings using an independent NSCLC cohort (cBioPortal[22]) and an in-house melanoma cohort treated with ICIs. Since only somatic sequencing data were available for the external NSCLC cohort, we evaluated the gene-level impact of the signature. Although survival differences did not reach statistical significance, likely due to biological distinctions between somatic and germline alterations, similar stratification trends were observed, with low-risk patients consistently showing superior survival curves. This suggests that the identified gene pathways may play a conserved role in modulating immunotherapy outcomes across contexts (Fig. 7c-d).

Discussion

Our study provides new evidence that germline variants, when analysed using advanced machine learning models, deliver valuable predictive information for immunotherapy response and outcomes in non-small cell lung cancer (NSCLC) [23]. Crucially, the robustness of these findings was supported by rigorous internal temporal validation and further explored in independent external cohorts, reinforcing the potential generalisability of the identified biomarkers. Unlike tumour-restricted or PD-L1-based biomarkers, inherited genomic variations appear to play an active role in determining therapeutic response, reflecting constitutional influences on host immune competence and tumour-immune interactions. Variants within genes such as *CCZ1B*, *SIGLEC11*, *PDE4E*, *ARHGAP22*, *PHLDB1*, *SLC6A16*, and *OR10H5* were consistently associated with progression-free survival (PFS), supporting the notion that the germline genome encodes mechanisms that modulate immune activation or resistance.

Through an agnostic approach, the machine learning algorithm identified several genes associated with immunotherapy response, some of which are linked to tumourigenic processes or have recently been implicated in pathways potentially regulating immunotherapy response in cancer. For example, *SIGLEC11* is a membrane receptor that negatively regulates the immune response by recognising and binding to sialoglycans on tumour cells, facilitating immune evasion and metastasis via the suppression of anti-tumour immunity [24]. Other relevant genes include *PDE4E*, a cAMP-specific phosphodiesterase linked to immunotherapy efficacy in LUAD [25], as well as *SLC6A16*, whose elevated expression in tumour cells has been shown to promote immunotherapy resistance by impairing CD8⁺ T cell function and reducing immune

infiltration [26]. Mechanistic and Prognostic Implications Beyond these specific gene functions, these findings collectively emphasise the utility of genome-wide, hypothesis-free approaches to uncover germline-mediated immunotherapy response mechanisms. However, other genes identified by this model have not yet been associated with tumour development or immune-related pathways. These findings may reflect statistical noise or the involvement of previously unrecognised germline mechanisms, warranting further functional investigations [27].

Among the favourable prognostic markers, *CCZ1B* showed the greatest survival benefit. The *CCZ1B* gene is involved in vesicular trafficking [28]; however, its specific role in boosting the cancer immune response remains to be elucidated. In contrast, variants in *OR10H5* were associated with resistance. Although *OR10H5* is canonically involved in olfaction, the identification of variants within this highly polymorphic gene family warrants caution because of the potential for mapping artefacts. While our stringent quality filtering minimises technical noise, we cannot exclude the possibility that this variant functions as a surrogate marker in a linkage disequilibrium with adjacent immune-regulatory loci. Future functional studies are essential to distinguish between the direct immunomodulatory role of *OR10H5* and the passenger effect. Notably, however, its expression has previously been implicated in cancer biology; for instance, *OR10H5* is part of a 19-gene prognostic signature in colorectal cancer, which correlates with improved patient survival [29]. Our data suggest a broader, context-dependent role in oncogenesis that warrants further investigation in NSCLC

Additionally, the *C22orf34* gene (a long non-coding RNA) was significantly associated with overall survival (OS) but not with PFS, suggesting that its prognostic relevance may be limited to long-term outcomes. Notably, *C22orf34* has been previously associated with an increased risk of drug-induced interstitial lung disease (DILD) in cancer patients [27], highlighting a potential link to treatment-related toxicity.

Beyond single-variant effects, the integrated germline–clinical score efficiently stratified patients according to PFS. The persistence of lymph node metastasis as an independent adverse factor underscores the multifactorial nature of resistance. Methodologically, the superior performance of XGBoost (AUC 0.845) over classical penalised regression approaches like LASSO (AUC 0.56–0.72) demonstrates the critical advantage of ensemble learning in capturing non-linear genotype–phenotype relationships. While traditional linear models often fail to account for the complex epistatic interactions inherent in genomic data, our ML approach effectively modelled these dependencies. This performance compares favourably with reported predictive metrics for standard biomarkers, such as PD-L1 alone, in real-world settings, suggesting that germline features capture a component of "missing heritability" not addressed by current tumour-only markers. Notably, predictive models using exomic data yielded slightly superior AUC compared to the entire variant set. This highlights the importance of feature selection, even when exonic variants directly encode protein-altering changes. The reduction of variables to the most informative subset improved model accuracy by excluding redundant or noisy features and minimising overfitting, a finding consistent with recent studies applying hybrid dimension reduction techniques to cancer genomics [30–32].

Nonetheless, this study had limitations. The single-country design and moderate sample size may limit generalisability and affect the effect estimates for rare variants. Despite strong internal validation and exploratory external validation cohorts, strictly matched germline replication studies in multi-ethnic NSCLC populations are needed to confirm these findings. A further limitation of our external validation is the reliance

on somatic sequencing data. Standard somatic variant calling pipelines often filter out germline signals as 'normal' background; therefore, our analysis likely captured somatic hits in the same genes rather than the original germline variants. However, the fact that somatic alterations in these loci also stratify risk supports the hypothesis that these genes are functional bottlenecks in tumour-immune interactions, regardless of the genomic origin (germline vs. somatic) of the alteration. Additionally, functional studies are required to causally connect these germline variations with immune-modulatory mechanisms.

Conclusion

Our findings establish a foundation for a comprehensive approach to discover immunotherapy biomarkers and stratify risk by integrating clinical and germline genetic data. This study illustrates how advanced AI-driven modelling can harness the predictive capabilities of inherited genetic variations, offering a pathway toward more personalised and precise immunotherapy strategies for lung cancer.

Abbreviations

NSCLC

Non-small cell lung cancer

LUAD

Lung adenocarcinoma

LUSC

Lung squamous cell carcinoma

ICI

Immune checkpoint inhibitor

PD-1

Programmed cell death protein 1

PD-L1

Programmed death-ligand 1

TMB

Tumor mutational burden

PFS

Progression-free survival

OS

Overall survival

CR

Complete response

PR

Partial response

SD

Stable disease

PD

Progressive disease

OR

Odds ratio
CI
Confidence Interval
DNA
Deoxyribonucleic Acid
RNA
Ribonucleic Acid
NGS
Next-generation sequencing
WES
Whole-exome sequencing
ML
Machine learning
AI
Artificial intelligence
EDTA
Ethylenediaminetetraacetic acid
ECOG
Eastern Cooperative Oncology Group
RECIST
Response Evaluation Criteria in Solid Tumours
QD
Quality by depth
VAF
Variant allele frequency
ACMG
American College of Medical Genetics and Genomics
FDR
False discovery rate
ROC
Receiver operating characteristic
AUC
Area under the curve
cAMP
Cyclic adenosine monophosphate

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the Declaration of Helsinki and approved by the Provincial Research Ethics Committee of Málaga (PEIBA), reference number 26-10-2017. Written informed consent was obtained from all participants included in the study.

Consent for publication

All authors have provided consent for publication.

Availability of data and materials

The datasets generated and/or analysed during the current study are available from the corresponding author upon reasonable request.

Competing interests

The authors declare no conflict of interest.

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Authors’ contributions

JO, AR, IB: conceptualization , AR , JLO, MRG, GPL: methodology and software, EPR, JCB , JZ, AC: clinical data curation, AGH, MGB, LCFO, BMG: experimental procedures, formal analysis, JO, AR: writing-original draft preparation, JO, AGR, AR: writing-review & editing, JO, ARD: supervision. All authors have read and approved the final manuscript.

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Figures

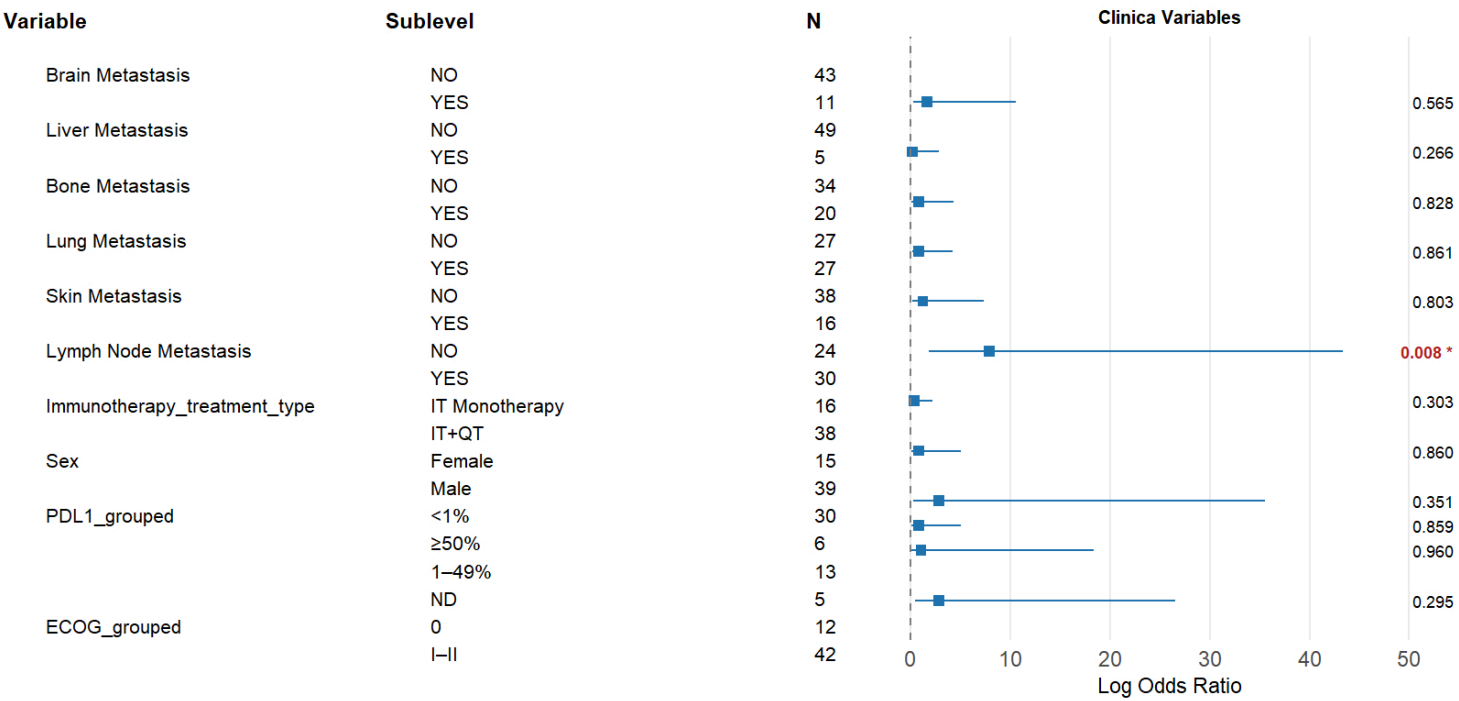


Figure 1

Forest Plot of Multivariable Predictors for ICI Response. Forest plot of adjusted odds ratios and 95% confidence intervals for clinical, pathological, and molecular features in the logistic regression model for the immunotherapy response. Only the presence of lymph node metastasis reached statistical significance (OR > 2, P = 0.008), whereas all other tested variables (brain, liver, bone, lung, treatment type, sex, PD-L1 expression, and cfDNA) showed no significant individual effects.

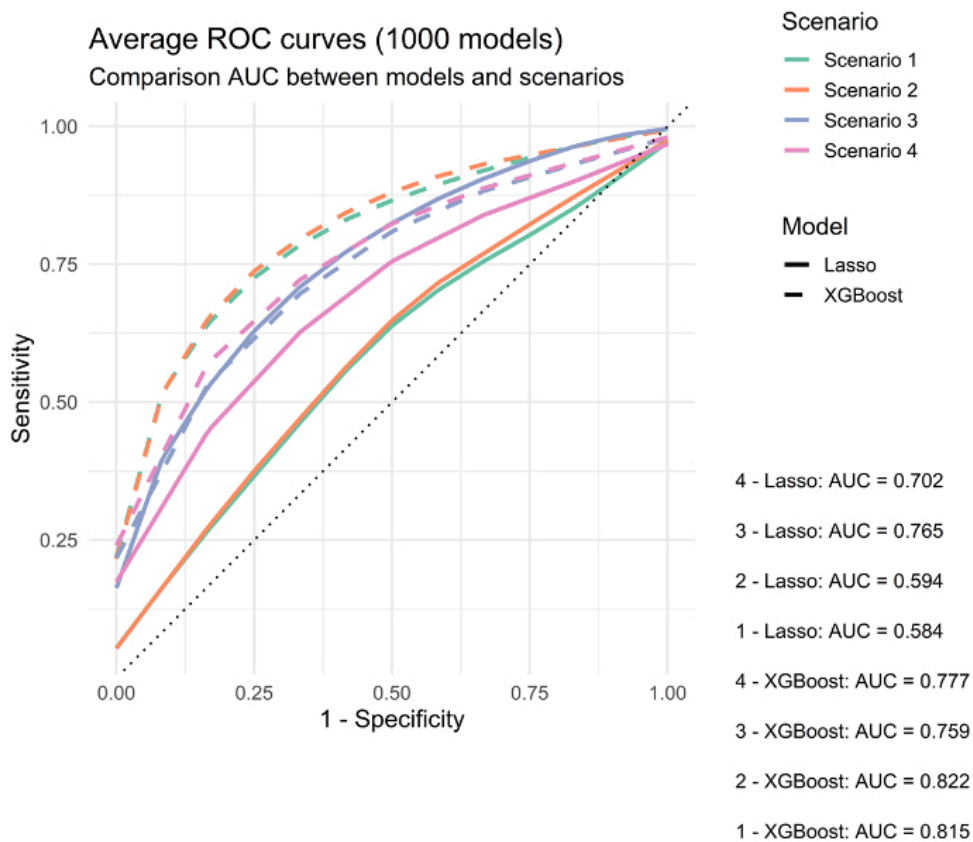
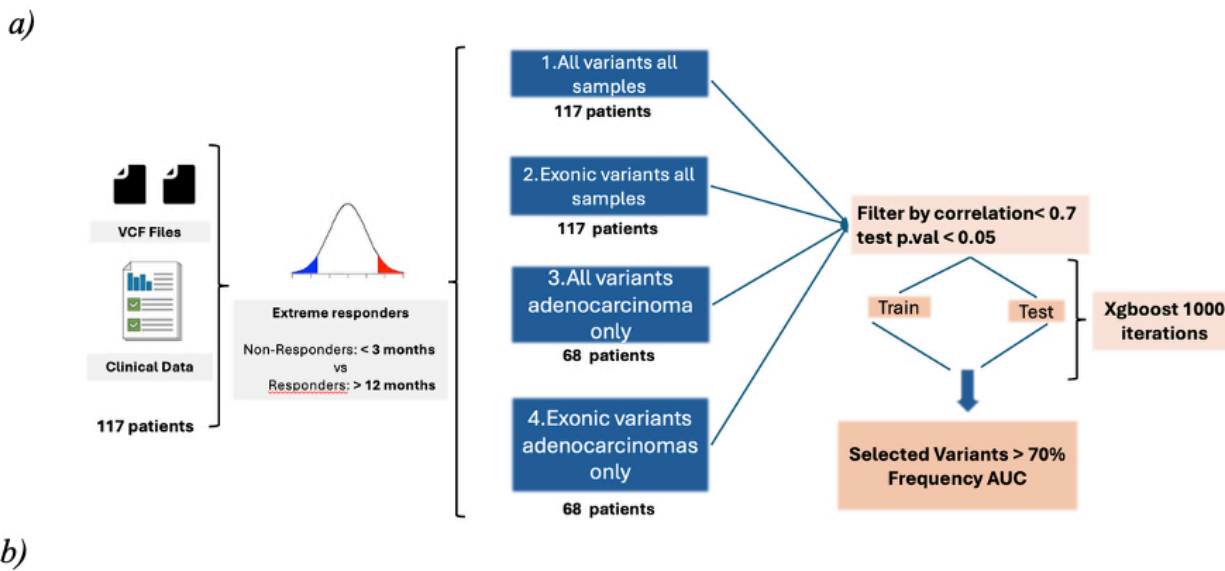


Figure 2

Workflow for Germline Variant Processing and Predictive Modelling. a) Schematic of the stepwise processing pipeline. Raw Whole-Exome Sequencing data were filtered for quality, aligned, and annotated to yield high-confidence germline variants. Feature selection involved removing highly correlated variants and prioritising those with significant clinical associations. b) Average ROC curves of the four analytical scenarios were modelled using the XGBoost and LASSO models.

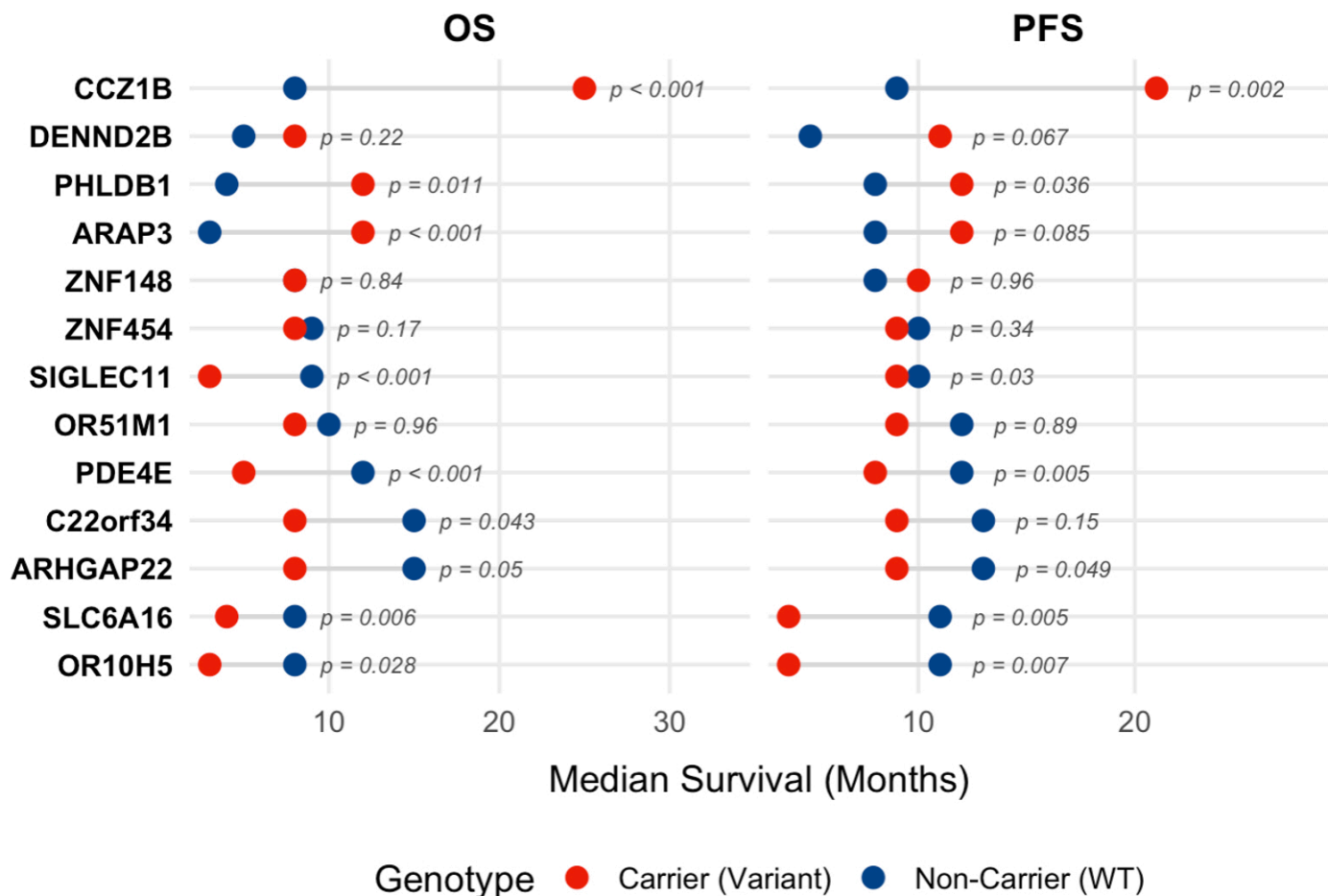


Figure 3

Impact of Germline Variants on Survival Outcomes. Dumbbell Plot comparing Median Overall Survival (OS, left panel) and Progression-Free Survival (PFS, right panel) between variant carriers (red) and non-carriers (blue). The analysis included the top germline variants prioritised by the machine learning model. P-values derived from the log-rank test are displayed for significant associations. Consistent with the multivariate analysis, variants in *CCZ1B*, and *PHLDB1*, were associated with extended survival, whereas *PDE4E*, *SLC6A16*, *OR10H5* and *SIGLEC11* were linked to significantly reduced survival times.

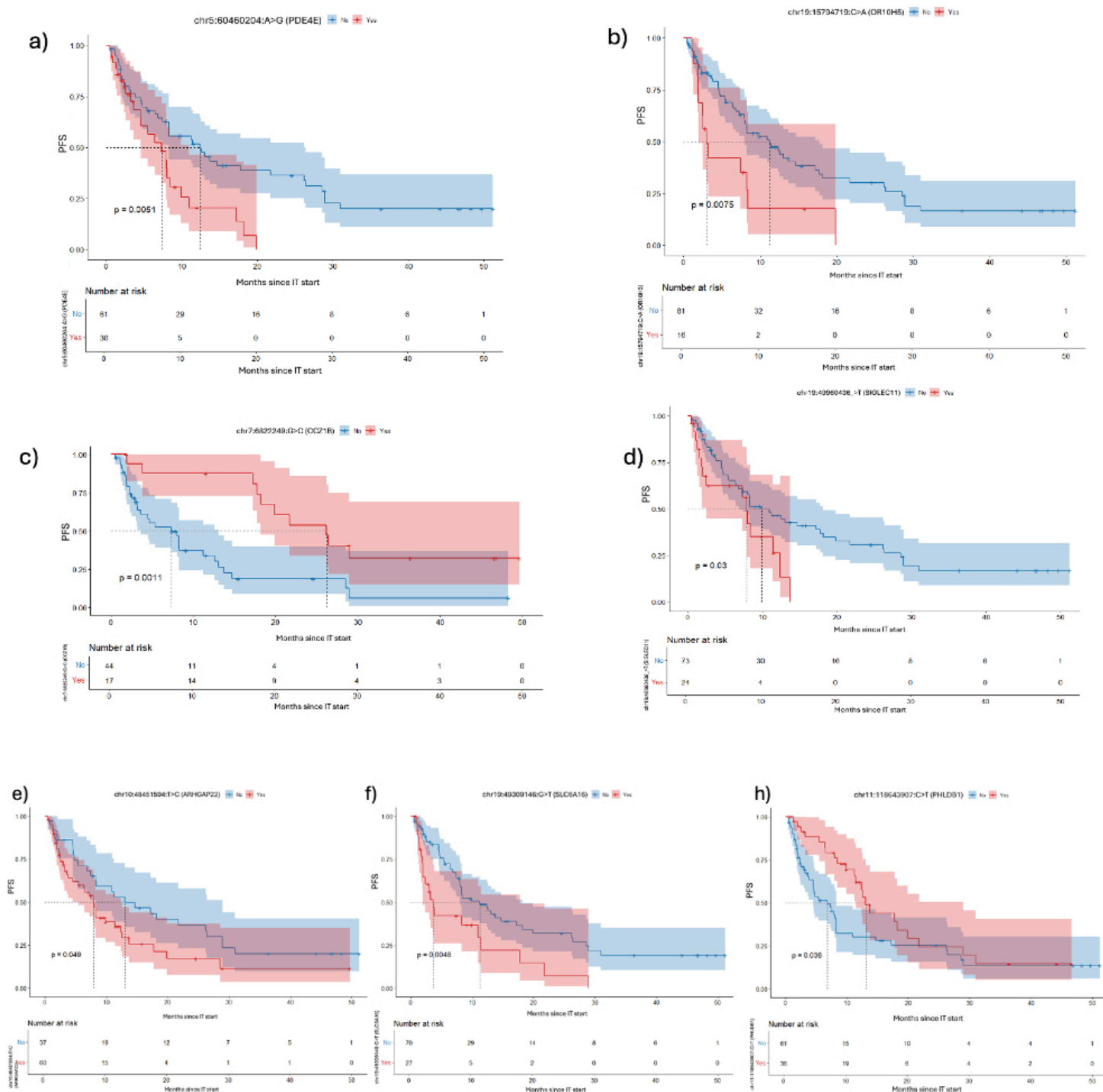


Figure 4

Impact of Germline Variants on Survival Outcomes. Dumbbell Plot comparing Median Overall Survival (OS, left panel) and Progression-Free Survival (PFS, right panel) between variant carriers (red) and non-carriers (blue). The analysis included the top germline variants prioritised by the machine learning model. P-values derived from the log-rank test are displayed for significant associations. Consistent with the multivariate analysis, variants in *CCZ1B*, and *PHLDB1*, were associated with extended survival, whereas *PDE4E*, *SLC6A16*, *OR10H5* and *SIGLEC11* were linked to significantly reduced survival times.

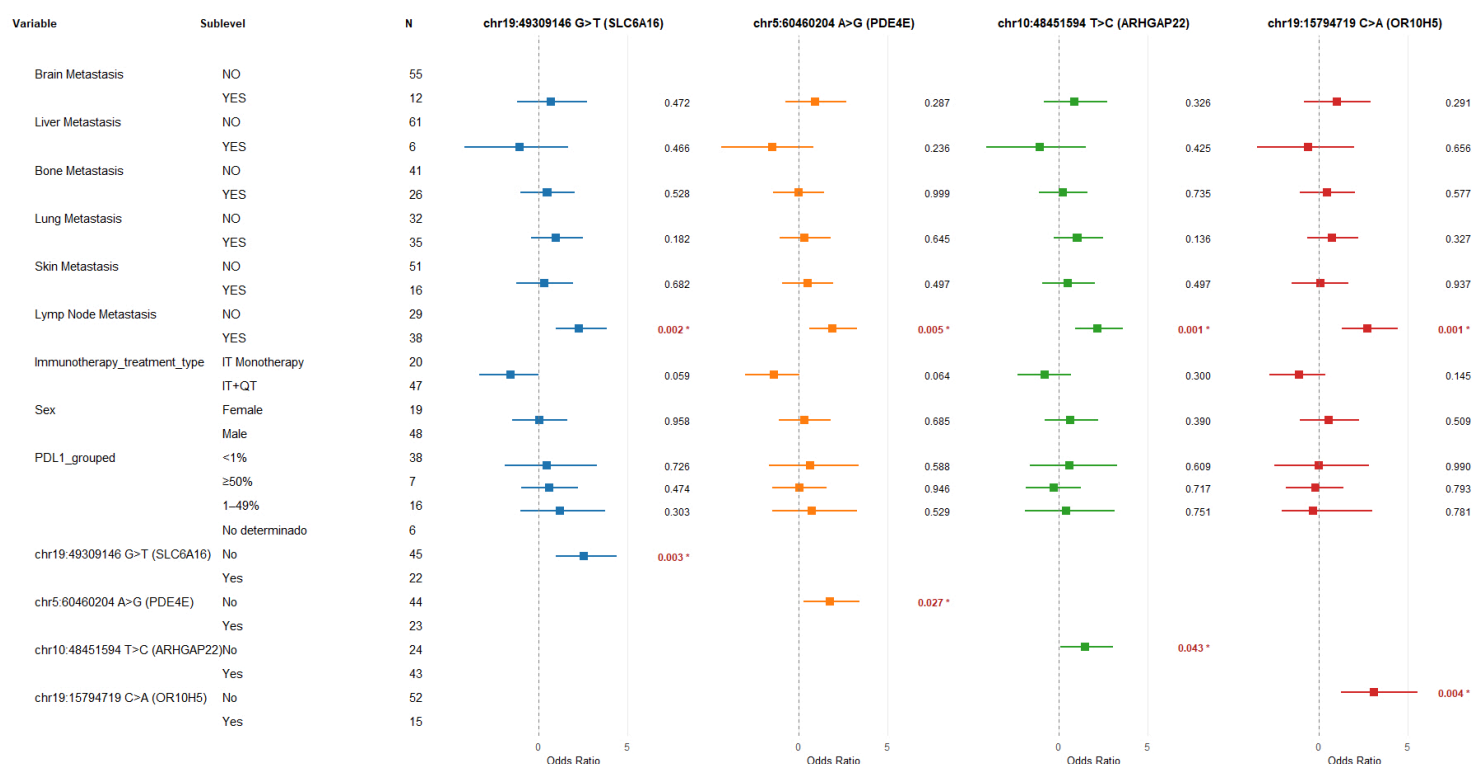


Figure 5

Multivariable Logistic Regression of Clinical and Germline Predictors of Immunotherapy Resistance. Forest plots displaying the independent association of clinical factors (metastatic sites, PD-L1 expression, ECOG status, treatment type) and specific germline variants with poor clinical response. Separate multivariable models were generated for each prioritised variant: *SLC6A16*, *PDE4E*, *ARHGAP22*, and *OR10H5*. Red squares indicate statistically significant associations ($P < 0.05$). In all models, lymph node metastasis and the presence of the indicated germline variants were consistently identified as independent predictors of treatment resistance ($OR > 1$).

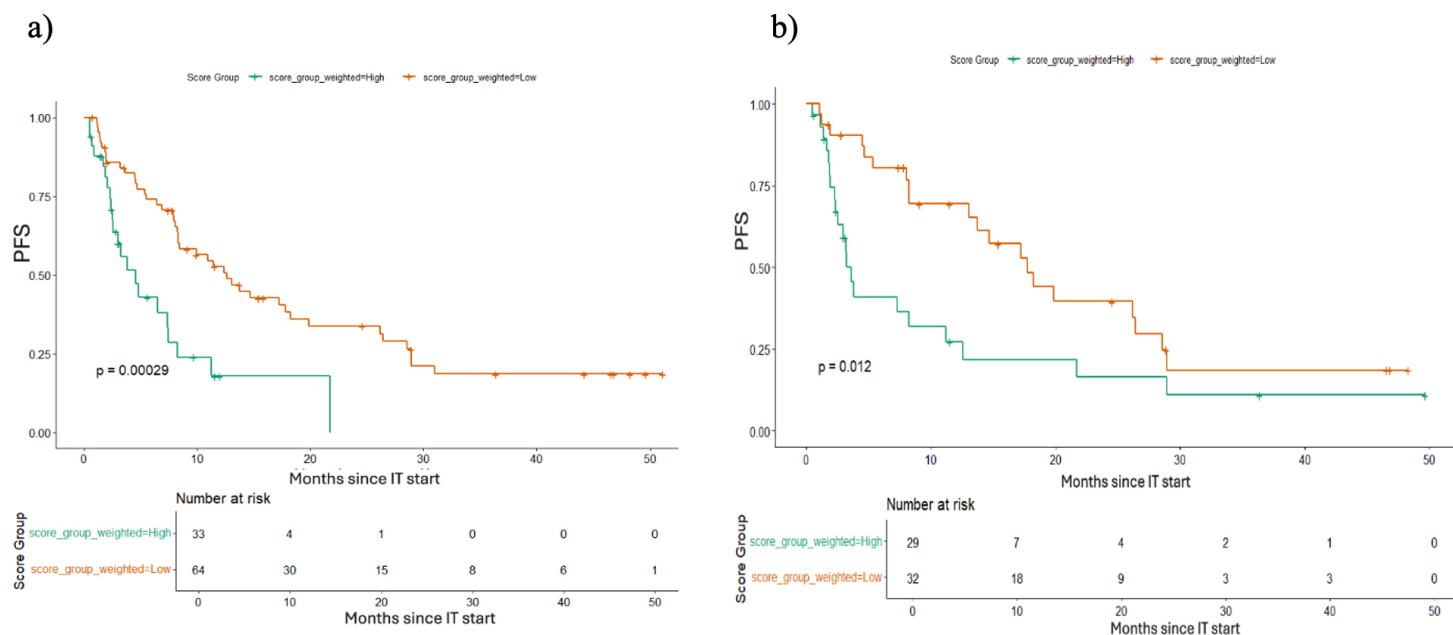


Figure 6

Clinical–Germline Risk Score Stratifies Progression-Free Survival. Kaplan–Meier survival curves for **(a)** the total patient cohort and **(b)** the lung adenocarcinoma (LUAD) subgroup. Patients were stratified into “low” (orange) and “high” (green) risk categories using a composite score that integrated the prioritised germline variants and lymph node metastasis status. Risk tables display the number of patients at risk at each time point in the study. The composite score significantly stratified outcomes in both the general and histological subgroups (log-rank $P < 0.001$ and $P = 0.012$, respectively).

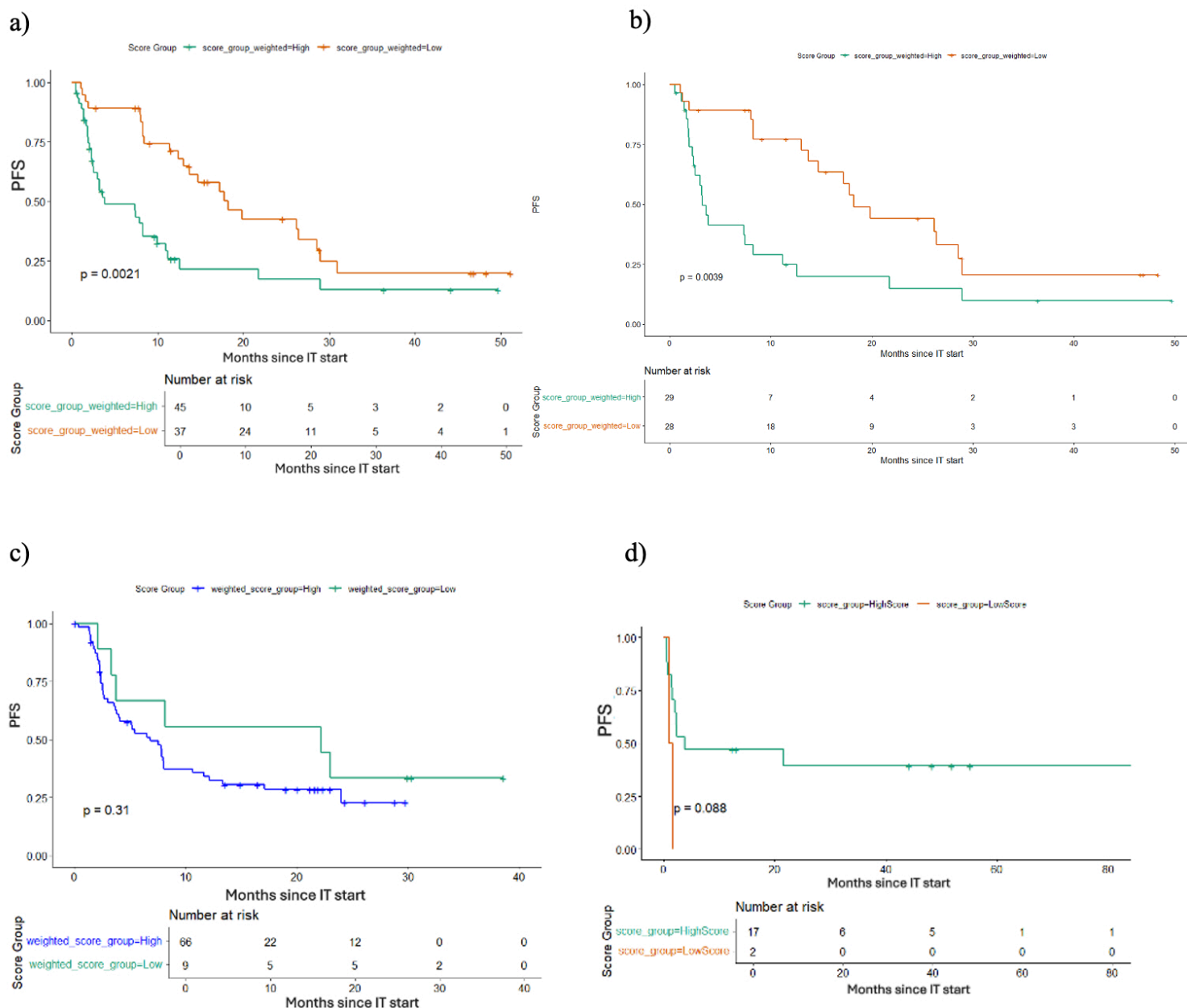


Figure 7

Internal and External Validation of the Risk Score. (a-b) Internal Temporal Validation: PFS stratification in the temporal hold-out validation set for (a) all subtypes and (b) LUAD patients, applying the original model cutoffs. (c-d) External Exploratory Validation: Kaplan-Meier curves for (c) an independent NSCLC cohort (cBioPortal, somatic data proxy) and (d) an independent melanoma cohort treated with ICIs. Although statistical significance was not reached in the external somatic/melanoma cohorts, the separation of curves followed the trend observed in the discovery cohort.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [GAXgboost.png](#)
- [SupplementaryMaterialtable1.docx](#)